MIL-C-60625 (MU)
30 NOVEMBER 1966

MILITARY SPECIFICATION

CASTINGS; DUCTILE (NODULAR) IRON FOR ARTILLERY, WARHEAD AND MORTAR AMMUNITION COMPONENTS

1. SCOPE

1.1 This specification covers ductile iron castings, alloyed and unalloyed, at all strength levels, intended for artillery, warhead and mortar ammunition components.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 — Ammunition and Special Weapons, General Specification For

MIL-P-10025 — Packaging, Packing and Marking for Interplant Shipment of Inert Ammunition Components, General Specification For

STANDARDS

FEDERAL

FED-STD-151 — Metal; Test Methods

MILITARY

MIL-STD-109 — Quality Assurance Terms and Definitions

MIL-STD-1167 — Ammunition Data Card

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

ASTM-A445-63T — Specification for Ferritic Ductile Iron Castings

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Preproduction.

FSC 1395

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:50Z
Check the source to verify that this is the current version before use.

3.1.1 *Sample castings.* Unless otherwise specified in the invitation for bids, contract, or order, prior to production, preproduction sample castings shall be furnished which will be poured from the metal representing the composition to be used in production. The castings shall be representative of the methods of molding, gating, risering, chilling, pouring and heat treatment to be used in production. The preproduction samples shall be subjected to all or any tests in this specification as specified in the invitation for bid, contract or order. No changes shall be made in the production of castings after acceptance of the preproduction samples without the approval of the procuring activity.

3.1.2 *Foundry procedure.* After acceptance of the sample castings, the producer shall submit his foundry process to the procuring activity. This procedure should show the following, as a minimum:

(a) Melting practice.

(b) Desulphurization practices.

(c) Inoculation procedure and specific alloy of inoculant.

(d) Post inoculation procedure.

(e) Pouring temperature (max, min).

(f) Allowable time to complete pour.

(g) Specific mean chemistry and limits and method of controlling chemistry.

(h) Molding medium.

(i) Pattern design.

(j) Core practice.

(k) Control of shakeout time.

(l) Heat treatments following casting.

(m) Final heat treatment procedure.

Any changes in foundry procedure shall require the re-submittal of sample castings to the appropriate agency for testing and approval or disapproval.

3.2 Chemical composition. The chemical composition of the castings shall meet the chemical composition ranges as shown in the Foundry Procedure (3.1.2), except that the carbon equivalent (C. E. = %C + 0.3 (%Si + %P)) shall be within the range 4.3 to 4.5, the maximum silicon content shall be 3.00 percent, and the maximum spread for the carbon range shall not exceed 0.20 percent carbon.

3.3 Microstructure.

3.3.1 *Nodularity.* At any position in the casting the graphite shall be classified as to type in accordance with the classification chart shown in figure 5 of ASTM–A445. The acceptable graphite formation shall contain a minimum of 90 percent types I and II graphite.

3.3.2 *Primary carbide.* The microstructure, at any location in the casting, shall contain not more than 5 percent primary carbide.

3.4 Mechanical properties. The mechanical properties shall be as specified on the drawing or as agreed upon between the foundry and the procuring activity.

3.5 Heat treatment. All castings shall be subjected to the heat treatment outlined in 3.1.2. Material in the as-cast condition shall not be acceptable under this specification. Various heat treatments consist of stress relieving, annealing (ferritizing or carbide solution) normalizing and austenitize, oil quench and tempering.

3.6 Soundness. The soundness of castings shall meet standards furnished or approved by the procuring agency using inspection methods specified in the item specification and by the procuring activity. The quality

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:50Z
Check the source to verify that this is the current version before use.

standard shall consist of a sample casting or castings, radiographs, photomicrographs, macrographs, other standards specified by the procuring activity, or a combination of these which shall clearly indicate the quality to be obtained.

3.7 Repair of castings. Defective casting shall not be plugged or welded.

3.8 Surface finish and condition.

3.8.1 *Machined surfaces.* Machined surfaces shall meet the surface finish requirements specified on the applicable drawings.

3.8.2 *As-cast surfaces.* As-cast surfaces on the finished part shall be as specified on the applicable drawing and specification for the end item, and shall be free of scale resulting from the casting process and subsequent heat treatment operations. Surface defects such as sandholes, gas holes, dross inclusions, core line impressions or any similar defect more severe than those permitted by standards provided or stipulated by the procuring activity shall be cause for rejection of the casting. Grinding to remove surface defects shall be permitted to minimum thickness requirements provided that thickness measurements are taken in each ground area to insure compliance with drawing requirements. Grinding shall be done in such a manner as to provide a smooth contour and shall have no sharp corners or edges.

3.9 Dimensions. Dimensions and dimensional tolerances of finished castings shall be as specified in the applicable drawing.

3.10 Identification markings. The identification mark of the manufacturer, pattern number, date and time of casting shall be cast into the surface of each casting using a format and location as approved by the procuring activity. If casting configuration or pattern design does not permit the inclusion of identification marks, methods of identifying castings as to date and time of casting shall be agreed upon between the foundry and procuring activity.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Quality assurance terms and definitions.* Reference shall be made to MIL–STD–109 to define quality assurance terms used.

4.1.2 *Inspection.* Inspection shall be in accordance with MIL–A–2550.

4.2 Inspection provisions.

4.2.1 *Lot.*

4.2.1.1 *Casting lot.* A casting lot shall consist of castings poured from the same inoculated ladle.

4.2.1.2 *Heat treatment lot.* A heat treatment lot shall consist of one or more casting lots not to exceed 4 hours of continuous foundry production or a maximum of 5,000 items, whichever is greater. All castings in a heat treatment lot shall be subjected to the same heat treatment cycle (temperature and time) in the same type of furnace equipment. The lot thus constituted shall be the basis for selection of sample castings to be used for the tension tests.

4.3 Inspection testing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:50Z
Check the source to verify that this is the current version before use.

4.3.1 *Chemical composition.* The chemical analysis of each casting lot shall be determined in compliance with foundry procedure (3.1.2) on a separately cast coupon, poured from the same inoculated ladle from which the casting lot was poured. Failure of the chemical analysis to comply with chemistry limits established by the foundry (3.1.2) and restrictions of 3.2, shall be cause for rejection of the casting lot.

4.3.2 *Microstructure.* A test coupon, the design of which shall be approved by the procuring activity, shall be cast at the end of each inoculated ladle to determine compliance with 3.3.1 and 3.3.2 when tested in accordance with 4.4.1. Casting lots rejected for failure to comply with 3.3.2, many, at the discretion of the foundry, be segregated into a separate heat treatment lot. The heat treatment lot shall then be heat treated in such a manner as to eliminate primary carbides. After heat treatment, two castings shall be selected at random, sectioned and examined for compliance with 3.3.2.

4.3.3 *Mechanical properties.*

4.3.3.1 *Hardness.* In order to insure uniformity of heat treatment and compliance with specified mechanical properties, each casting, after final heat treatment, shall be subjected to a hardness test with a Brinell hardness tester at a location of the casting specified or approved by the procuring agency.

4.3.3.1.1 *Alternate hardness testing.* A direct reading Brinell hardness testing machine may be substituted for the Brinelloscope method, provided the same diameter ball is used and the same load is applied as is used in standard Brinell testing, and provided that at least 2,000 castings are tested by both methods to establish a correlation between the two methods. After a correlation has been established, the diameter of the indentation in every tenth casting shall be measured with the Brinelloscope to assure continued correlation. Methods other than the indentation type may be used provided correlation with the Brinelloscope method can be made to the satisfaction and approval of the procuring agency.

4.3.3.2 *Tension test.* Two castings from each heat treatment lot representing the hardest and softest castings, as determined by the hardness test outlined in 4.3.3.1 and 4.3.3.1.1, shall be subjected to the tension test. Two specimens from each casting shall be prepared as specified in 4.4.2.2 and tested. If all specimens comply with the requirements of 3.4 the hardness range for the lot will have been established. Failure of one or more tension test specimens to comply with the requirements shall be cause for rejection of the heat treatment lot represented by the specimen.

4.3.3.3 *Re-test.* When one or more tension test specimens fail to comply with the requirements of 3.4 and do not qualify for retest under 4.4.2.2.3 a new hardness range shall be established. The rejected lot may, at the discretion of the foundry, be re-heat-treated prior to establishing a new hardness range. The tension test, in compliance with 4.3.3.2, shall be conducted to determine the acceptability of the new hardness range. When the new hardness range has been established, all castings whose hardness values fall outside this range shall be rejected.

4.3.4 *Soundness.* Each projectile body casting or any other casting that is subject to being loaded with high explosives or chemical filler, shall be tested in accordance with 4.4.3 or as prescribed by the procuring activity. Failure of any casting to comply with 3.6 shall be cause for rejection of the casting. Inspection and sampling on other ammunition components shall be as specified by the procuring activity. When inspection is performed on a sampling basis, the lot shall be rejected if the inspection criteria has been reached or exceeded.

4.3.5 *Identification markings.* At least one casting from each heat treatment lot shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:50Z
Check the source to verify that this is the current version before use.

be visually examined for legibility of markings and for proper location to determine compliance with 3.10.

4.3.6 *Surface finish, as-cast.* Each casting shall be visually examined to determine compliance with 3.8.2. Failure of casting to comply with 3.8.2 will be cause for rejection of the individual casting.

4.4 Test methods and procedures.

4.4.1 *Microstructure.* A metallographic sample shall be prepared from the test coupon to determine compliance with 3.3.1. The polished specimen shall represent the full cross-section of the test coupon. To determine compliance with 3.3.1 the specimen shall be examined uneteched, at 100 magnification, at several locations, at a distance from the as-cast surface where microstructure is not affected by surface condition. To determine compliance with 3.3.2 the specimen prepared for 3.3.1 shall be etched with nital or other suitable etching reagent to reveal matrix microstructure. The specimen will be examined at 100 magnification and traversed from one as-cast surface to the opposite surface to determine presence of primary carbide.

4.4.2 *Mechanical properties.*

4.4.2.1 *Hardness.* The Brinell hardness test shall be performed in accordance with procedures prescribed in FED–STD–151. Due to the nature of ductile iron, indenter balls of less than 10-millimeter diameter shall not be used.

4.4.2.2 *Tension test.* From each sample selected in accordance with 4.3.3.2 two tension test specimens shall be taken longitudinally from each side of the sample 180° apart, or as prescribed by the procuring activity.

4.4.2.2.1 *Tension test specimen.* The specimen shall be the largest obtainable standard round specimen prescribed by FED–STD–151 (types R1 through R5) which the size of the casting and location of the specimen will permit. If the size or geometry of the casting does not permit a standard round specimen, a specimen specified or approved by the procuring activity shall be used.

4.4.2.2.2 *Yield strength, tensile strength and elongation.* Yield strength shall be determined by the offset method as defined in FED–STD–151. The offset shall be 0.1 percent (0.001 inch per inch of gage length). Tensile strength and elongation on each specimen shall be determined by utilizing the criteria for these tests prescribed by FED–STD–151.

4.4.2.2.3 *Replacement test specimens.* If a tension test specimen fails to meet the requirements of 3.4 due to an identifiable defect in the tensile specimen which is acceptable under the provisions of 3.6 and 4.3.4 or due to a fracture occurring under those conditions allowing for re-tests for elongation in FED–STD–151, another specimen shall be machined from the same casting and tested. In the instance of failure due to an identifiable metal defect, only one replacement specimen shall be permitted, and failure of the replacement specimen to comply with the requirements of 3.4 shall be cause for rejection of the represented lot. Results of all tests, original and replacement tests, shall be recorded and specimens which failed to meet the requirements due to the above conditions shall be identified and retained by the contractor and made available to the procuring activity for examination upon request.

4.4.3 *Soundness.* For those applications where 100 percent inspection is required, castings will be inspected by ultrasonics, or, if the geometry of the casting does not permit the use of ultrasonics, some form of radiography and magnetic particle inspection may be used, if approved by the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:50Z
Check the source to verify that this is the current version before use.

4.4.4 *Chemical analysis.* The chemical analysis shall be determined by the direct combustion method (carbon), wet analysis or spectroanalysis methods recognized by ASTM and approved by the procuring activity. The separately cast coupon of 4.3.1 shall be crushed or otherwise divided. Drillings shall not be permitted as sample for the total carbon determination.

5. PREPARATION FOR SHIPMENT

5.1 Packing.

5.1.1 *Level A.* Not applicable.

5.1.2 *Level B.* Not applicable.

5.1.3 *Level C.* Castings shall be packed in accordance with the requirement of MIL–P–10025.

5.2 Labeling and marking. Labeling and marking shall be in accordance with the requirements of MIL–P–10025.

6. NOTES

6.1 Intended use. Ductile iron castings purchased to this specification are intended for use in artillery, mortar, and warhead ammunition.

6.2 Tension test specimens. Separately cast test coupons for tension test specimens are not always representative of the mechanical properties present in the casting and, therefore, shall not be used in tension tests for determining compliance with mechanical property requirements unless specifically authorized by the procuring activity.

6.3 Government owned tooling. Due to the technical aspects involved, the mold patterns and core boxes shall be Government owned tooling.

6.4 Data cards. When specified by the contracting officer, data cards shall be required as specified in MIL–STD–1167.

Custodian:

Army — MU

Preparing activity:

Army — MU

Project No. 1395–A026

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:50Z
Check the source to verify that this is the current version before use.